IoT-Based HVAC Temperature and Humidity Monitoring System

1. Project Overview

The **IoT-Based HVAC Temperature and Humidity Monitoring System** is designed to continuously monitor the environmental conditions inside a building, laboratory, office, industry, or HVAC-controlled room. The system uses a **DHT22 temperature and humidity sensor** to measure real-time temperature and relative humidity. An **ESP8266 Wi-Fi microcontroller** processes the sensor data and transmits it to an IoT cloud platform. The collected information can be monitored remotely using a mobile phone or computer. If the temperature or humidity exceeds the predefined safe range, the system can activate a **buzzer, warning LED, or HVAC control relay**. This improves indoor comfort, energy efficiency, equipment protection, and environmental management.

2. Problem Statement

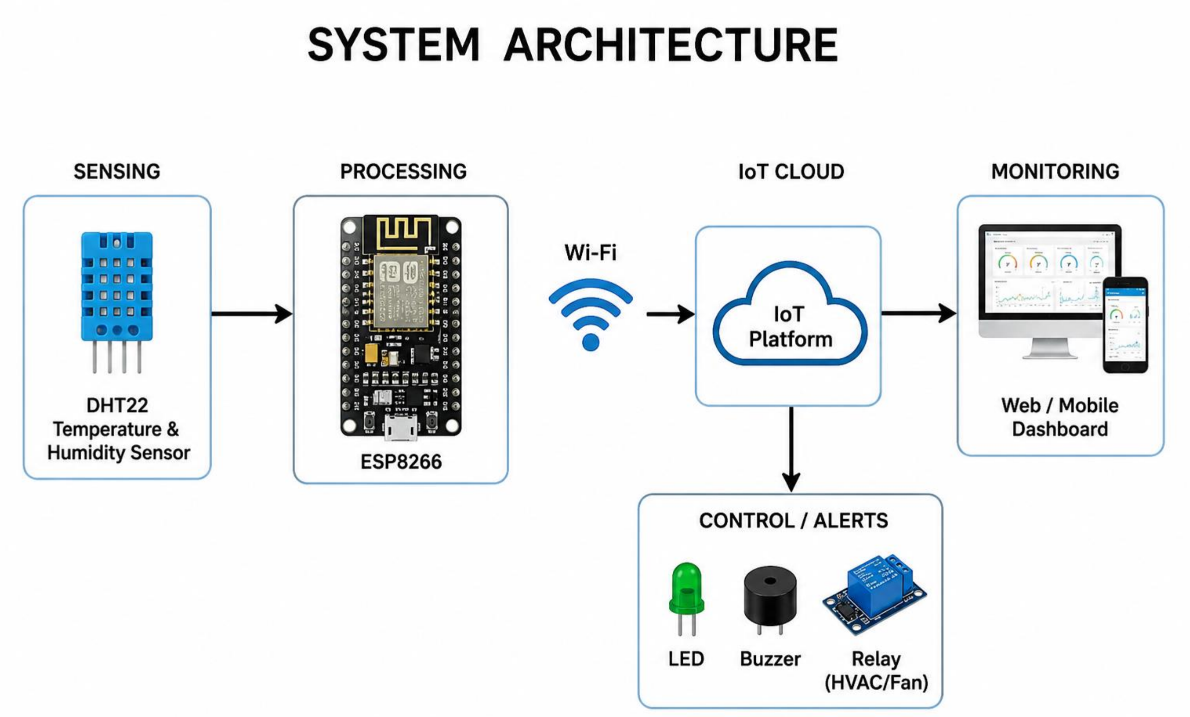
Traditional HVAC systems generally operate according to manually configured temperature settings and may not provide continuous monitoring of humidity and environmental conditions. Improper temperature and humidity levels can cause discomfort, excessive energy consumption, condensation, equipment damage, and poor indoor environmental quality. In large buildings and industrial environments, manually checking these parameters at different locations is time-consuming and inefficient. There is therefore a need for an intelligent monitoring system that can continuously measure temperature and humidity, provide remote access to the data, generate alerts during abnormal conditions, and assist in automatic HVAC control.

3. Proposed Solution

The proposed system uses a **DHT22 sensor, ESP8266 Wi-Fi module, display unit, buzzer, LEDs, and relay module** to create an IoT-enabled HVAC monitoring system. The DHT22 sensor continuously measures temperature and humidity and sends the readings to the ESP8266. The ESP8266 processes the data and uploads it to an IoT cloud platform through Wi-Fi. Users can monitor the environmental parameters remotely using a smartphone or computer. When the measured values exceed predefined limits, the system provides an alert and can control a connected HVAC demonstration load through a relay. This approach provides continuous monitoring, faster response, reduced manual inspection, and better energy management.

4. System Architecture

The system consists of five major sections: sensing, processing, communication, monitoring, and control.



The sensor provides the environmental data, while the ESP8266 acts as the main processing and communication unit. The IoT platform stores and displays the data remotely, and the relay provides an interface for HVAC control.

5. Circuit Design

The DHT22 sensor is connected to the ESP8266 to measure temperature and humidity. The sensor receives 3.3 V supply from the controller and its data pin is connected to the selected GPIO pin. An OLED display can be connected through the I²C interface to locally display temperature and humidity values.

A green LED indicates that the environmental conditions are within the desired range, while a red LED and buzzer indicate an abnormal condition. A relay module is connected to the ESP8266 to demonstrate automatic control of a fan or HVAC load.

6. Circuit Table

Component	Pin/Terminal	ESP8266 Connection	Function
DHT22	VCC	3.3V	Sensor power
DHT22	GND	GND	Ground
DHT22	DATA	D4 / GPIO2	Temperature & humidity data
OLED/LCD	VCC	3.3V	Display power
OLED/LCD	GND	GND	Ground
OLED/LCD	SDA	D2 / GPIO4	I²C data
OLED/LCD	SCL	D1 / GPIO5	I²C clock
Green LED	Anode	D5 / GPIO14	Normal condition
Red LED	Anode	D6 / GPIO12	Warning condition
Buzzer	+	D7 / GPIO13	Alarm indication
Relay	IN	D8 / GPIO15	HVAC/fan control
Relay	VCC	5V	Relay power
Relay	GND	GND	Common ground

A suitable transistor/driver or relay module should be used when required by the relay input specification. For an actual HVAC unit, proper electrical isolation and a correctly rated contactor/relay are essential.

7. Algorithm

Start the system and initialize the ESP8266, DHT22 sensor, LEDs, buzzer, and relay.

Connect the ESP8266 to the available Wi-Fi network.

Read the temperature and humidity values from the DHT22 sensor.

Process and display the measured temperature and humidity values.

Send the sensor data to the IoT cloud platform through Wi-Fi.

Compare the measured values with the predefined temperature and humidity limits.

If the values are within the safe range, turn ON the green LED and keep the HVAC system in normal operation.

If the values exceed the safe range, turn ON the red LED and buzzer and activate the relay for HVAC control.

Update the monitoring data continuously on the IoT dashboard.

Repeat the process continuously for real-time HVAC monitoring.

Algorithm Steps

1. Start the system.
2. Initialize the ESP8266 and DHT22 sensor.
3. Connect the ESP8266 to the available Wi-Fi network.
4. Read temperature and humidity values from the DHT22 sensor.
5. Check whether the sensor readings are valid.
6. Display the readings on the local display.
7. Send the data to the IoT cloud platform.
8. Compare the readings with predefined limits.
9. If the conditions are normal, activate the green LED.
10. If the values exceed the limits, activate the red LED and buzzer.
11. Control the HVAC/fan through the relay according to the programmed condition.
12. Repeat the monitoring process continuously.

8. Coding

```
#include <DHT.h>


// =====================================================

// PIN CONNECTIONS

// =====================================================
```

```
#define DHT_PIN     D7
#define DHT_TYPE    DHT22

#define RELAY_PIN   D5
#define BUZZER_PIN  D6

// ====================================================
// DHT22 SENSOR
// ====================================================

DHT dht(DHT_PIN, DHT_TYPE);

// ====================================================
// TEMPERATURE SETTING
// ====================================================

// HVAC turns ON when temperature reaches 30°C
float temperatureLimit = 30.0;

// ====================================================
// BUZZER TYPE
// ====================================================

// Set true if your 3-pin buzzer is ACTIVE LOW
// Set false if your 3-pin buzzer is ACTIVE HIGH

#define BUZZER_ACTIVE_LOW true

// ====================================================
```

```
// BUZZER ON
// ==========================================================

void buzzerON()
{
  if (BUZZER_ACTIVE_LOW)
  {
    digitalWrite(BUZZER_PIN, LOW);
  }
  else
  {
    digitalWrite(BUZZER_PIN, HIGH);
  }
}

// ==========================================================
// BUZZER OFF
// ==========================================================

void buzzerOFF()
{
  if (BUZZER_ACTIVE_LOW)
  {
    digitalWrite(BUZZER_PIN, HIGH);
  }
  else
  {
    digitalWrite(BUZZER_PIN, LOW);
  }
```

```
}

// ==========================================================

// SETUP

// ==========================================================

void setup()
{
  // Start Serial Monitor
  Serial.begin(115200);

  // Start DHT22
  dht.begin();

  // Relay
  pinMode(RELAY_PIN, OUTPUT);

  // 3-pin buzzer
  pinMode(BUZZER_PIN, OUTPUT);

  // Start with HVAC OFF
  digitalWrite(RELAY_PIN, HIGH);

  // Start with buzzer OFF
  buzzerOFF();

  delay(2000);

  Serial.println();
```

```
  Serial.println("========================================");
  Serial.println(" IoT HVAC TEMPERATURE & HUMIDITY");
  Serial.println(" MONITORING SYSTEM");
  Serial.println("========================================");
  Serial.println("DHT22      : READY");
  Serial.println("Relay      : READY");
  Serial.println("Buzzer     : READY");
  Serial.println("System     : READY");
  Serial.println();
}

// ======================================================
// MAIN LOOP
// ======================================================

void loop()
{
  // ====================================================
  // READ DHT22
  // ====================================================

  float temperature = dht.readTemperature();
  float humidity = dht.readHumidity();

  // ====================================================
  // CHECK SENSOR
  // ====================================================

  if (isnan(temperature) || isnan(humidity))
```

```
{
  Serial.println("-------------------------------------");
  Serial.println("DHT22 SENSOR ERROR!");
  Serial.println("Check DHT22 connections.");

  // Safety: HVAC OFF
  digitalWrite(RELAY_PIN, HIGH);

  // Buzzer warning
  buzzerON();
  delay(300);
  buzzerOFF();

  Serial.println("-------------------------------------");

  delay(2000);

  return;
}

// ===================================================
// DISPLAY VALUES ON SERIAL MONITOR
// ===================================================

Serial.println("-------------------------------------");

Serial.print("Temperature : ");
Serial.print(temperature, 1);
Serial.println(" °C");
```

```
Serial.print("Humidity    : ");
Serial.print(humidity, 1);
Serial.println(" %");

// ====================================================
// HVAC CONTROL
// ====================================================

if (temperature >= temperatureLimit)
{
  // Temperature is HIGH
  digitalWrite(RELAY_PIN, LOW);

  // Buzzer ON
  buzzerON();

  Serial.println("Temperature : HIGH");
  Serial.println("HVAC        : ON");
  Serial.println("Relay       : ON");
  Serial.println("Buzzer      : ON");
}
else
{
  // Temperature is NORMAL
  digitalWrite(RELAY_PIN, HIGH);

  // Buzzer OFF
  buzzerOFF();
```

```
    Serial.println("Temperature : NORMAL");
    Serial.println("HVAC        : OFF");
    Serial.println("Relay       : OFF");
    Serial.println("Buzzer      : OFF");
  }

  Serial.println("------------------------------------");
  Serial.println();

  // Wait 2 seconds
  delay(2000);
}
```

9. System Working

When the system is powered ON, the ESP8266 initializes the DHT22 sensor and establishes a Wi-Fi connection. The DHT22 continuously senses the temperature and relative humidity of the HVAC environment. The ESP8266 receives the sensor readings and processes them according to the programmed limits.

The measured values can be displayed locally and transmitted to an IoT platform for remote monitoring. During normal environmental conditions, the green LED remains ON and the HVAC system operates normally. If the temperature or humidity exceeds the defined threshold, the ESP8266 activates the warning LED and buzzer and can switch the relay to control a fan or HVAC load. The monitoring process is repeated continuously, allowing the system to respond quickly to environmental changes.

10. Bill of Materials

S. No	Component	Quantity	Approx. Purpose
1	ESP8266 Node MCU	1	Main controller & Wi-Fi
2	DHT22 Sensor	1	Temperature & humidity sensing
3	OLED/LCD Display	1	Local data display

S. No	Component	Quantity	Approx. Purpose
4	1-Channel Relay Module	1	HVAC/fan control
5	Green LED	1	Normal indication
6	Red LED	1	Warning indication
7	Buzzer	1	Alarm indication
8	Resistors	2–3	LED protection
9	Breadboard	1	Prototype assembly
10	Jumper Wires	As required	Connections
11	5V USB Power Supply	1	Power source
12	Computer/Mobile	1	IoT monitoring

11. Prototype Implementation

The prototype can be constructed on a breadboard by connecting the DHT22 sensor, ESP8266, LEDs, buzzer, display, and relay module according to the circuit table. The ESP8266 is programmed using the Arduino IDE and connected to a Wi-Fi network. The DHT22 is positioned inside the test environment so that it can accurately sense temperature and humidity.

For demonstration, a small DC fan can be connected through the relay instead of directly connecting a high-voltage HVAC unit. Different environmental conditions can then be simulated by changing the surrounding temperature or humidity. The sensor readings are observed through the serial monitor and IoT dashboard, while the LEDs, buzzer, and fan demonstrate the automatic response of the system.

12. Results & Performance Analysis

The developed prototype provides continuous monitoring of temperature and humidity and successfully demonstrates IoT-based environmental monitoring. The ESP8266 can collect sensor readings and transmit them through Wi-Fi for remote observation. The warning mechanism provides immediate indication when the environmental conditions exceed the programmed limits.

The system reduces the need for manual monitoring and can support better HVAC management. The use of automatic control can help prevent unnecessary HVAC operation and improve energy utilization. The prototype is also scalable because additional sensors can be incorporated for monitoring multiple rooms or zones.

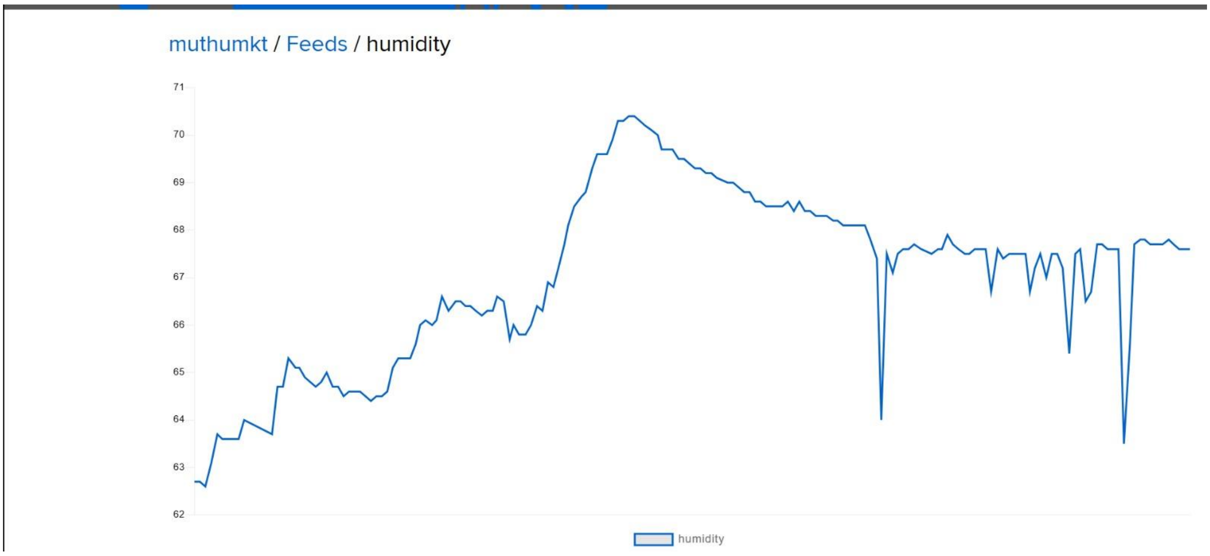
Expected Performance

Parameter	Expected Result
Temperature monitoring	Continuous
Humidity monitoring	Continuous
Remote monitoring	Wi-Fi/IoT enabled
Abnormal condition alert	LED + Buzzer
Automatic control	Relay based
Data accessibility	Mobile/Computer
Response	Near real-time
System operation	Continuous
Manual monitoring	Reduced
Energy management	Improved

Conclusion

The **IoT-Based HVAC Temperature and Humidity Monitoring System** provides an effective method for real-time environmental monitoring and intelligent HVAC management. By combining sensors, ESP8266 Wi-Fi connectivity, IoT monitoring, and automatic control, the system can improve comfort, safety, equipment protection, and energy efficiency. The prototype can be further enhanced by adding **multiple HVAC zones, cloud data logging, mobile notifications, energy-consumption monitoring, and AI-based predictive HVAC control**.

OUTPUT:




```
HVAC        : ON
Relay       : ON
Buzzer      : ON
--------------------------------------

--------------------------------------
Temperature : 716.8 °C
Humidity    : 1817.9 %
Temperature : HIGH
HVAC        : ON
Relay       : ON
Buzzer      : ON
--------------------------------------

--------------------------------------
Temperature : 716.8 °C
Humidity    : 1817.9 %
Temperature : HIGH
HVAC        : ON
Relay       : ON
Buzzer      : ON
--------------------------------------

--------------------------------------
Temperature : 716.8 °C
Humidity    : 1817.9 %
Temperature : HIGH
HVAC        : ON
Relay       : ON
Buzzer      : ON
--------------------------------------

--------------------------------------
Temperature : 716.8 °C
Humidity    : 1817.8 %
Temperature : HIGH
HVAC        : ON
Relay       : ON
Buzzer      : ON
--------------------------------------
```